Solution

ELECTRICITY -01

Class 10 - Science

1.

(c) Neutral

Explanation: The metallic conductor is electrically neutral. When a potential difference is set up across the metallic conductor, the loosely bound electrons in the metal move with a drift velocity.

2.

(d) 10^{-3} A and 10^{-6} A respectively

Explanation: μA = 10^{-6} a

mA = 10^{-3} a

3.

(d) 10^{20}

Explanation: We have current I = 1 A and time t = 16s

We know $I = \frac{Q}{t}$ or, $1. A = \frac{Q}{16s}$

or, Q = 16 C

Charge contained in 1 electron = 16 $\times 10^{-19} C$

So, 16 C charge is contained in the following number of electrons:

= $\frac{1}{1.6 \times 10^{-18}} \times 16 = 10^{20}$

4.

(c) Ampere

Explanation: Ampere

5.

(d) 0.1 A

Explanation: Last cout = $\frac{Range}{Total\ divisions} = \frac{2}{20} = \frac{1}{10}$ = 0.1 A

6. **(a)** $\frac{\text{Work done}}{\text{Current} \times \text{Time}}$

Explanation: Voltage is equal to the work done per unit of a charge against a static electric field to move the charge between two points. A voltage may represent either a source of energy (electromotive force), or lost, used, or stored energy (potential drop).

Hence, voltage is represented as Voltage = $\frac{\text{Work done}}{\text{Current} \times \text{Time}}$

7.

(b) Volt

Explanation: Volt is the unit used for measuring the potential difference.

8.

(c) (a) - (iv), (b) - (i), (c) - (ii), (d) - (iii)

Explanation: (a) - (iv), (b) - (i), (c) - (ii), (d) - (iii)

9. **(a)** Potential difference

Explanation: Potential difference is the measure of the work done in moving a unit charge across two points in an electric circuit.

10.

(b) 1.6 V

Explanation: Voltage drop in arm containing 2Ω and 3Ω is 4 V. Let voltage drop across each resistor be 2x and 3x.

$\therefore$ 2x + 3x = 4 $\Rightarrow$ x = $\frac{4}{5}$

$\therefore$ Voltage across 2Ω is 2x

i.e., 2 $\times \frac{4}{5}$ = 1.6 V

11. **(a)** Voltmeter

Explanation: Voltmeter is the device that is used for measuring potential difference.

12.

(c) 0.025 V

Explanation: Least count = $\frac{Range}{Total\ divisions}$ = $\frac{0.5-0}{20}$ = $\frac{0.5}{20}$ = $\frac{5}{200}$ = 0.025 V

13.

(c) 10 Joule

Explanation: Work done = V × C = 10 × 1 = 10 joule

14.

(c) heating effect of current

Explanation: The heating effect of current is the working principle for an electric fuse.

15.

(c) 20A

Explanation: Given,

Charge moved, Q = 20C

Time taken, t = 1s

To find,

Current, I = ?

We know that the current is calculated as follows:

I = $\frac{Q}{t}$

Substituting the values, we get

I = $\frac{20}{1}$ = 20A

Therefore, 20A current is required to move a charge of 20C in 1s time.

16.

(c) Very high resistance

Explanation: A voltmeter is used to measure the potential difference in the circuit. It is always connected in parallel because it has very high resistance.

17.

(c) 0.5 A

Explanation: Given,

Charge moved = 300 C

Time taken = 10 mins

Current = ?

I = $\frac{Q}{t}$

Substituting, we get I = 0.5 A

18. **(a)** 40 C

Explanation: Given,

Current, I = 4A

Time taken, t = 10s

To find,

Charge moved, Q = ?

We know that,

I = $\frac{Q}{t}$

Substituting the values in the above equation, we get

4 = $\frac{Q}{10}$

Q = 4 $\times$ 10 = 40 C

Therefore, 40C charge is moved in 10s when the current flowing is 4 A.

19.

(c) 31.25 $\times$ 10^{18}C

Explanation: Given,

Current, I = 5A

Time taken, t = 1s
We know that,
I = $\frac{Q}{t}$
Q = I $\times$ t = 5 $\times$ 1 = 5C
Number of electrons present in 1.6 $\times$ 10^{19}C = 1
Number of electrons present in 5C = $\frac{(5)}{(1.6\times10^{19})}$ = 31.25 $\times$ 10^{18}C

20.
(b) 0.5 A
Explanation: Given,
Charge moved, Q = 20C
Time taken, t = 40s
To find,
Current, I = ?
We know that,
I = $\frac{Q}{t}$
Substituting the values in the above equation we get
I = $\frac{20}{40}$ = 0.5A
Therefore, 0.5A is required for moving 20C charge in 40s.

21.
(d) Resistance
Explanation: Slope of V-I graph represents the resistance in the conductor.
Resistance R = $\frac{V}{I}$

22. **(a)** Nichrom
Explanation: Nichrom

23.
(b) 9.2 A
Explanation: 9.2 A

24.
(d) straight line
Explanation: If a graph drawn between the potential difference and current the graph is found to be a straight line passing through the origin. From the graph, we see that Potential difference (V) and current (I) directly proportional to one another.

25.
(c) 125 volt
Explanation: First case,
Given,
Current, I = 0.02A
Voltage, V = 10 volt
We know that,
V = IR
10 = 0.02 $\times$ R
R = $\frac{10}{0.02}$ = 500 Ohms
Second case,
Current, I = 250 $\times$ 10^{-3}A
Resistance, R = 500 Ohm
V = IR
V = 250 $\times$ 10^{-3} $\times$ 500 = 125 volt

26. **(a)** All of these
Explanation: All of these

27. **(a)** 6.25×10^{-18}

Explanation: Given,

Charge moved = 1.6×10^{-19} coulombs

Current, I = 1A

Time taken, t = 1s

We know that,

$I = \frac{Q}{t}$

$1 = \frac{Q}{1}$

Q = 1C

We also know that when the charge is 1.6×10^{-19} coulombs, the number of electrons present is 1

Therefore, when a charge of 1 coulomb is present, the number of electrons is given as:

$\frac{(1)}{(1.6 \times 10^{-19})} = 0.625 \times 10^{-19} = 6.25 \times 10^{-18}$

28.

(d) 40 V

Explanation: Given,

Resistance, R = 20 Ohms

Current, I = 2 amp

We know that,

V = IR

Therefore, $V = 2 \times 20 = 40V$

29.

(b) 18×10^{6} J

Explanation: Given,

I = 5A

R = 100 ohm

T = 2 hours

We know that,

Energy consumed = $P \times T = I^2RT$

Substituting the values,

Energy consumed = 5kWh

1kWh = 3.6×10^{6} J

Therefore, 5kWh = $5 \times 3.6 \times 10^{6}$ J = 18×10^{6} J

30.

(c) Directly proportional to the potential difference

Explanation: According to Ohm's law, the current passing through a conductor is directly proportional to the potential difference between two points.

31.

(c) Current and potential difference

Explanation: Current and potential difference

32.

(b) While plotting this graph, the temperature remains constant.

Explanation: The ratio of potential difference applied to the wire and current passing through it is a constant. Ohm’s law is illustrated by this graph. Hence, temperature must be constant while calculating values.

33.

(d) 24 Ω

Explanation: 24 Ω

34.

(b) 2 Ω

Explanation: 2 Ω

35.

(b) A semiconductor

Explanation: A semiconductor

36.

(d) Rheostat

Explanation: Rheostat

37.

(b) 2 A

Explanation: 2 A

38.

(b) 3

Explanation: 3

39.

(c) Ohm

Explanation: Ohm

40.

(b) Potential difference (V) decreases

Explanation: Potential difference (V) decreases

41. **(a)** $\frac{\text{1 Joule}}{\text{1 Coulomb}}$

Explanation: $\frac{\text{1 Joule}}{\text{1 Coulomb}}$

42.

(d) 484 ohm

Explanation: Given,

V = 220V

P = 100W

R = ?

We know that,

R = $\frac{V2}{P}$

R = 484 ohm

43.

(c) 6.25×10^{18}

Explanation: One electron contains a charge of 1.6×10^{-19} C. Therefore, one coulomb = $\frac{1}{1.6} \times 10^{-19}$ electrons = 6.25×10^{18} electrons.

44.

(c) 800 volt

Explanation: Given,

Current, I = 200 mA = 0.2A

Resistance, R = 4kΩ = 4×10^3 Ohm = 4000 Ohm

We know that,

V = IR

V = 0.2×4000

V = 800 volt

45. **(a)** 50 C

Explanation: Given,

Current, I = 5A

Time taken, t = 10s

Charge moved, Q = ?

I = $\frac{Q}{t}$

Q = I $\times$ t
Q = 50 C

46.

(b) 6.25 $\times$ 10^{18} electrons

Explanation: There are 6.25 $\times$ 10^{18} electrons in one-coulomb charge.

47.

(c) 0.6 Ohms
Explanation: Given,
Current, I = 5A
Potential difference, V = 3V
We know that,
V = IR
Therefore,
3 = 5 $\times$ R
R = $\frac{3}{5}$ = 0.6 Ohms

48.

(b) Four times
Explanation: Heat produced in a conductor
H = $\frac{V^2}{R}$
If resistance is reduced to one fourth, then
H' = $\frac{V^2}{(R/4)}$ or H' = 4$\left[\frac{V^2}{R}\right]$
$\therefore$ H' = 4H

49. **(a)** 5.4 $\times$ 10^6 J
Explanation:
According to Joule's law of heating, the heat produced in a resistor is directly proportional to the square of the current for a given resistance, directly proportional to the resistance for a given current and directly proportional to the time for which the current flows through the resistor.
Given P = VI = 1500 W and t = 1 hour
$\therefore$ Heat produced H = P $\times$ t = 1500 W $\times$ 1 hour = (1500 W) $\times$ (60 $\times$ 60 sec) = 5.4 $\times$ 10^6 J.

50. **(a)** 1 A
Explanation: In a series connection, the current through each device remains the same. Therefore, the current through the 40 W bulb will also be 1 A.

51.

(d) 4.8 amp
Explanation: First case:
Given,
Current, I = 2.4amp
Voltage, V = 120V
We know that,
V = IR
120 = 2.4 $\times$ R
R = $\frac{120}{2.4}$ = 50 Ohm
Second case:
Given,
Voltage, V = 240 volt
Resistance, r = 50 Ohm
We know that,
V = IR
240 = I $\times$ 50
I = 4.8 amp

52. **(a)** Fuse wire

Explanation: We know that in a series connection, if one circuit gets damaged, then the other circuits connected to it will also get damaged. The wire connected necessarily in series is fuse wire, to carry the current in a closed path.

53.

(c) 5 A

Explanation: Current flowing through the kettle can be calculated as follows: p = $V \times I$

or, $1000\text{W} = 220\text{V} \times 1$

or $I = \frac{1000\text{W}}{220\text{V}} = 4.54\text{A}$

So, the required rating of fuse wire = 5A

54. **(a)** four times

Explanation: The heat produced in an electric kettle becomes four times the actual value when the current is doubled.

55.

(c) 92 J/s

Explanation: Given,

V = 230V

I = 0.4 amp

Rate at which electrical energy is transferred = Power

P = VI

P = 92watt = 92 J/s

56. **(a)** nichrome

Explanation: The elements of electric heating devices are made using nichrome.

57.

(b) 20 J

Explanation: The total resistance of combination = $2\Omega + 4\Omega = 6\Omega$

Current through the circuit can be calculated as follows: $I = \frac{V}{R} = \frac{6V}{6\Omega} = 1A$

Heat dissipation by 4 can be calculated a follows: $H = I^2Rt = (1.A)^2 \times 4\Omega \times 5s = 20J$

58.

(b) 15 A

Explanation: 15 A current is required for the safe working of an electric kettle that has a supply of 230 V.

59.

(b) 300%

Explanation:

- Power dissipation will be given by P = I^2R where 'I' is the current and 'R' is the resistance at the constant temperature.
- New current, I' = I + (100 percent of I) or I' = 2I
- New power dissipation, P' = $(2I)^2$R or P' = $4I^2$R
- Increase in power dissipation, P' - P = ($4I^2$R) - (I^2R)
- Percent increase in power dissipation = $\frac{P' - P}{p} \times$ 100% = 300%

60.

(c) 345.6 J

Explanation: Given,

R = 25 Ω

V = 12 V

T = 60s

H = ?

V = IR

12 = 25I

I = 0.48 amp

We know that,

$H = I^2RT$
$H = 345.6$ J

61. **(a)** 1250 J/s
Explanation: Given,
R = 200 ohms
I = 2.5 amp
T = 1s
We know that,
$H = I^2RT$
H = 1250 J/s

62. **(a)** one-fourth
Explanation: When the current flowing through a fixed resistor is half, the heat produced in the resistor will become four times the actual values.

63.
(c) Rupees 24
Explanation: It is given that the electrical energy cost is Rupees $\frac{4}{kWh}$
Time duration is 3 hours
The power of the electric heater is 2kW
Using the given information we can calculate the amount for 3 hours as Rupees 24

64.
(b) Kilowatt-hour
Explanation: Kilowatt-hour is the commercial unit of energy.

65.
(c) 20 V
Explanation: Given,
H = 100 J
T = 1s
R = 4 ohms
We know that,
$H = I^2RT$
$100 = I^2 \times 4 \times 1$
$\frac{100}{4} = I^2$
I = 5 amp
V = IR
$V = 5 \times 4$
V = 20 V

66.
(d) Parallel to the line wire
Explanation: Parallel to the line wire

67. **(a)** 59.4 kWh
Explanation: Case 1:
Power, P_1 = 60W
Number, n_1 = 2
Time for use, T_1 = 4 hours everyday
Energy consumed, $E_1 = n_1 \times P_1 \times T_1$
$E_1 = 2 \times 60 \times 4$ = 480 watt-hour = 0.48kWh
Therefore, energy consumed for 30 days = 30×0.48 = 14.4 watt-hour
Case 2:
Power, P_2 = 100W

Number, $n_2 = 3$

Time for use $T_2 = 5$ hours everyday

Energy consumed, $E_2 = n_2 \times P_2 \times T_2$

$E_2 = 3 \times 100 \times 5 = 1500 = 1.5$kWh

Therefore, energy consumed for 30 days = $30 \times 1.5 = 45$kWh

Therefore, overall energy consumed = 14.4 + 45 = 59.4kWh

68.

(c) $x \times z \times y^2$

Explanation: $x \times z \times y^2$

69.

(b) two times

Explanation: If V is constant, then H is inversely proportional to R because $H = V^2 t/R$. H will therefore double if R does.

70. **(a)** 600 kW h

Explanation: V = 260 V, P = 2 kW = 2000 W,Time for which air conditioner is switched on(t) = 10 hours x 30 = 300 hours

Electrical energy consumed in 30 days

$= p \times t = 2000 \times 300$

= 600000 W h = 600 kW h

71.

(b) Joule

Explanation: The SI unit of energy is joule.

72.

(c) 1800 J/s

Explanation: Given,

R = 8 ohm

I = 15 amp

T = 1s

We know that,

$H = I^2RT$

H = 1800 J/s

73.

(b) Thick and short

Explanation: When a fuse wire is thick the electrons available are more which is not a suitable characteristic of fuse wire.